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(54) COOLING MEDIUM CIRCULATION APPARATUS FOR VEHICLE

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F16K 5/04 (2006.01)
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CPC ***B60H 1/00485*** (2013.01); ***F16K 5/0407*** (2013.01); ***F16K 5/0471*** (2013.01); ***F16K 11/0853*** (2013.01); ***F16K 27/065*** (2013.01)

(58) **Field of Classification Search**
CPC F16K 11/0853; F16K 11/0856; B60H 1/00485
See application file for complete search history.

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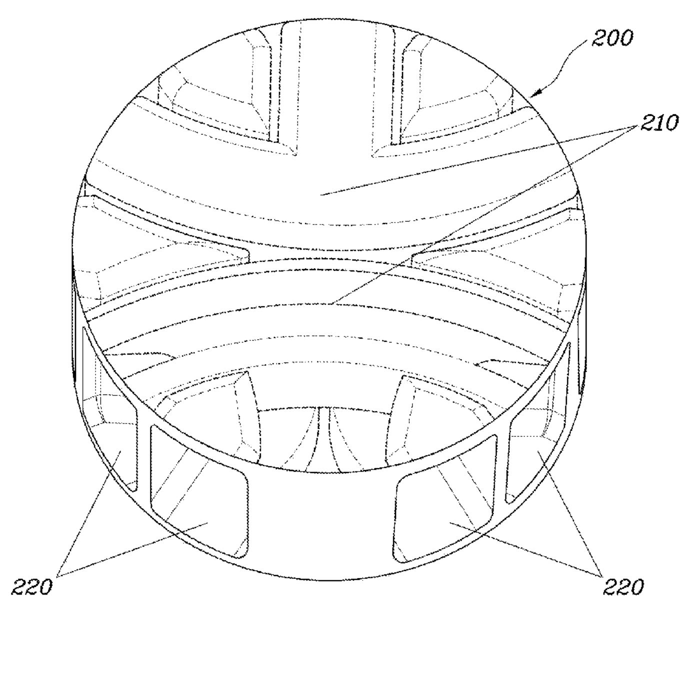
(57) **ABSTRACT**

A cooling medium circulation apparatus for a vehicle includes: a housing having a valve mounting device with an internal space and a plurality of ports connected to a plurality of thermal management components; and a valve body rotatably mounted in the valve mounting device, partitioned into a plurality of layers vertically, with a plurality of flow paths formed in each layer, and with a plurality of circulation holes formed on outer sides corresponding to each of the flow paths and selectively matching the ports, the circulation holes elongated vertically to have a height greater than a height of the flow paths. The cooling medium circulation apparatus may be installed in the vehicle in a compact manner, while eliminating the need for multiple peripheral pipes, and facilitating the circulation of coolant across various cooling system components.

20 Claims, 12 Drawing Sheets



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F16K 11/085	(2006.01)
F16K 27/06	(2006.01)

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FIG. 1

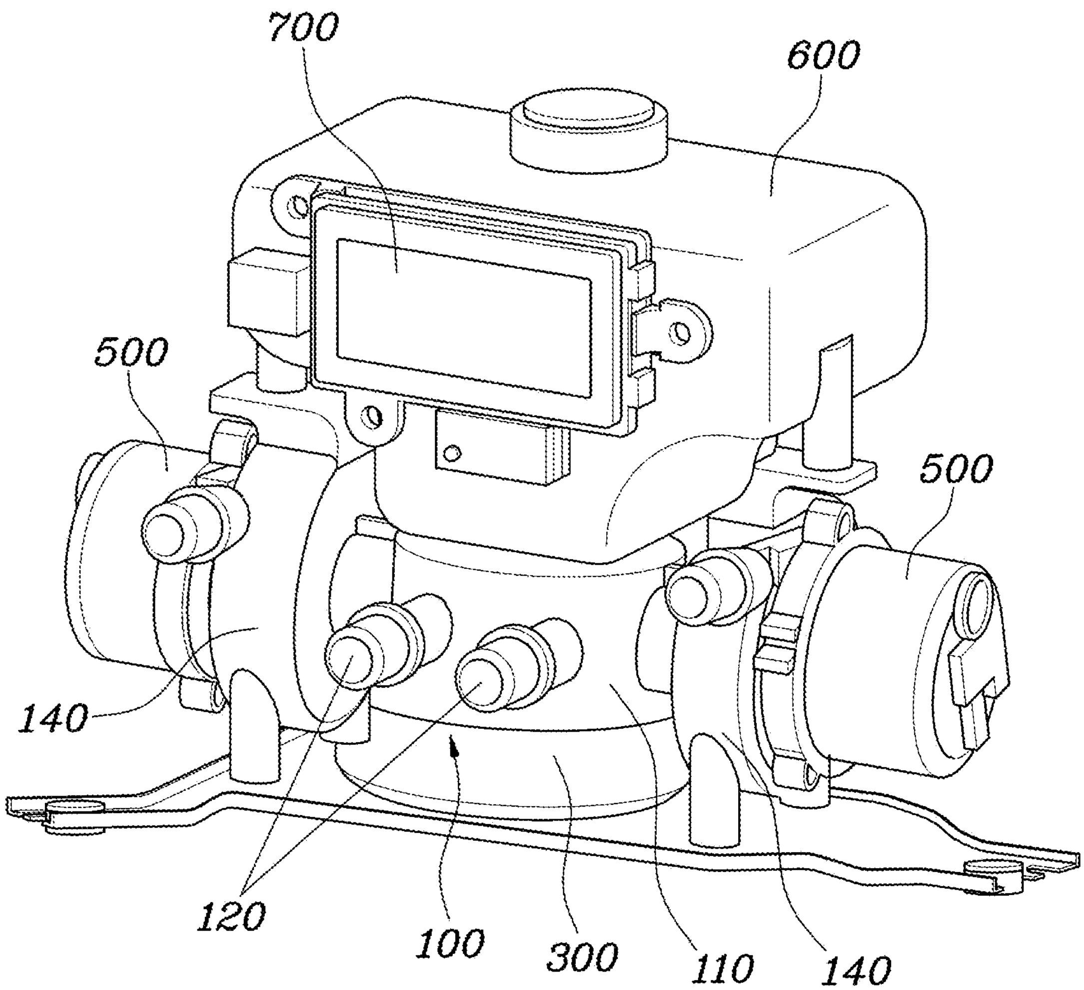


FIG. 2

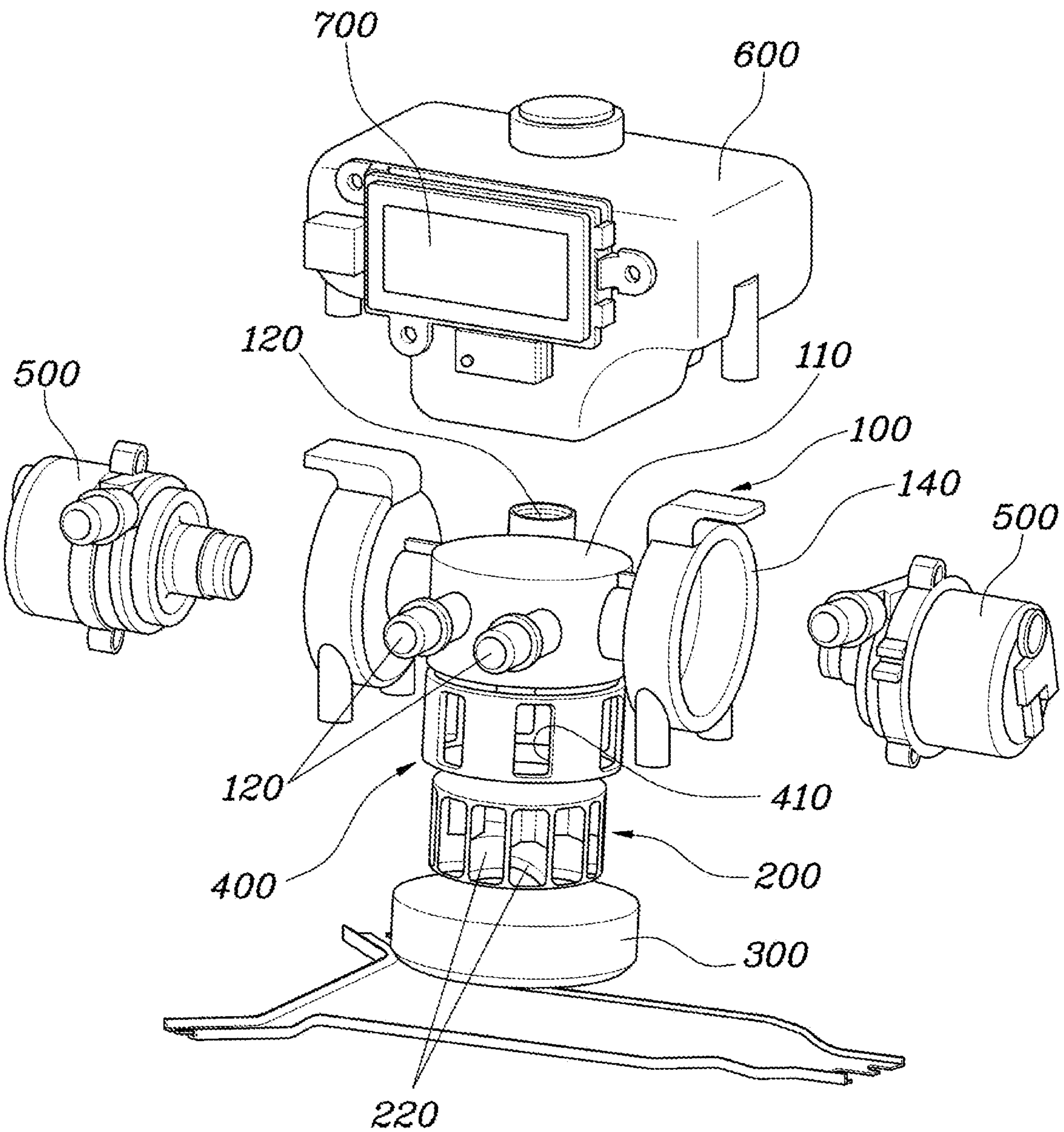


FIG. 3

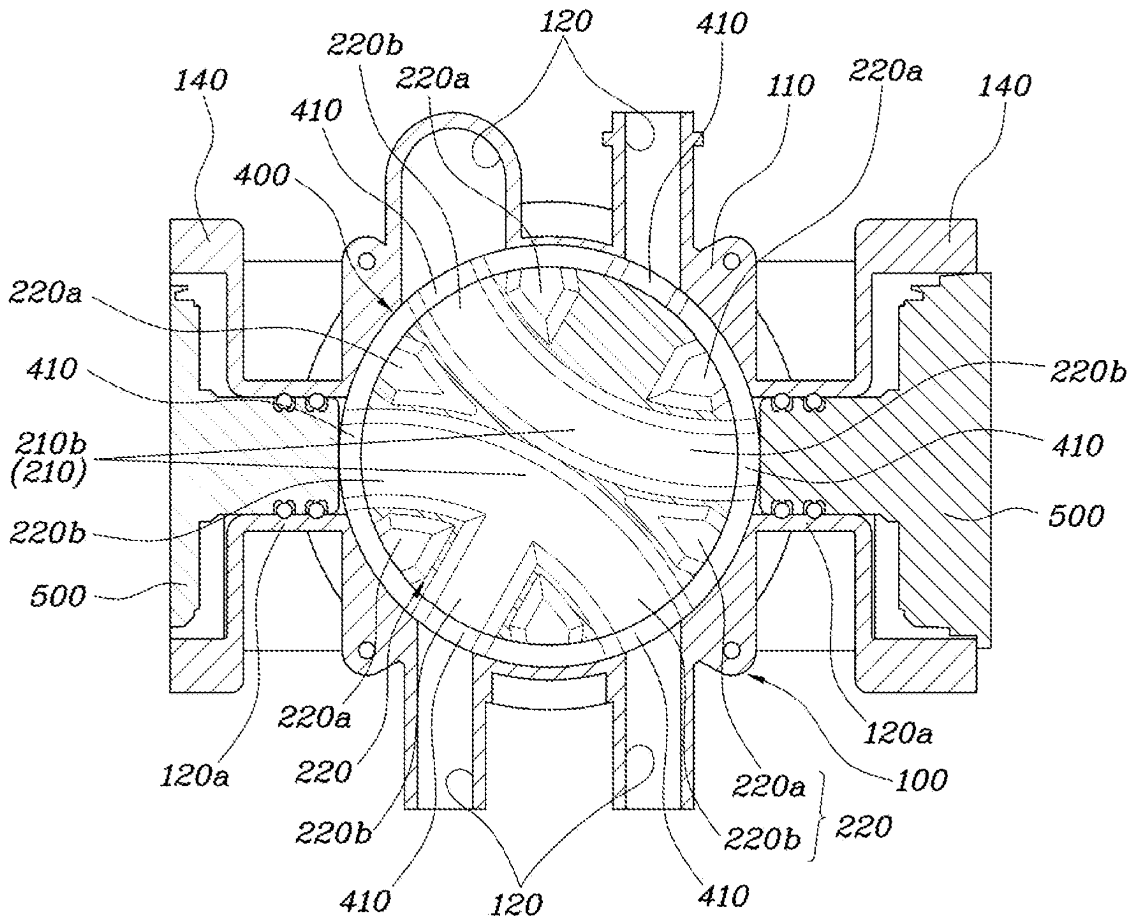

120
410
220b
220a
140
410
110
220a
140
400
220a
410
220b
410
210b
(210)
500
220b
500
220a
120a
220
120a
100
120a
220a
220
220b
220b
220
410
120
410

FIG. 4

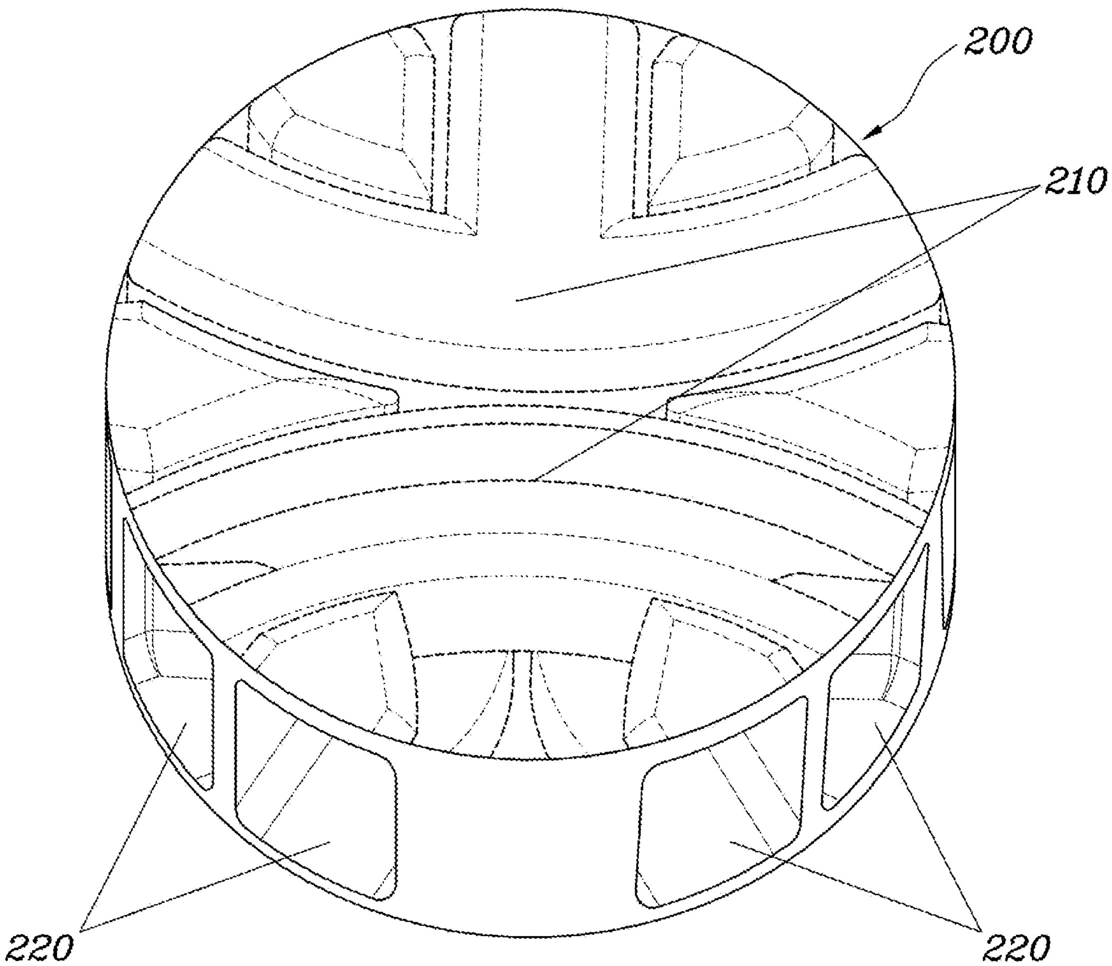


FIG. 5

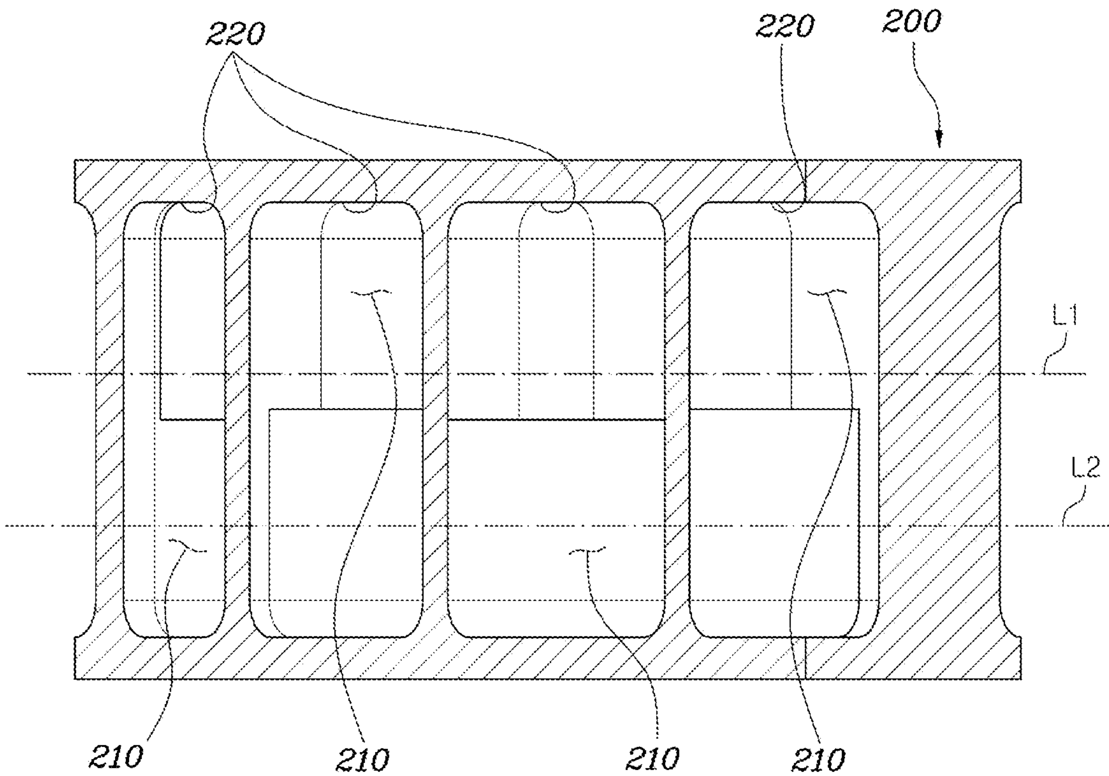


FIG. 6

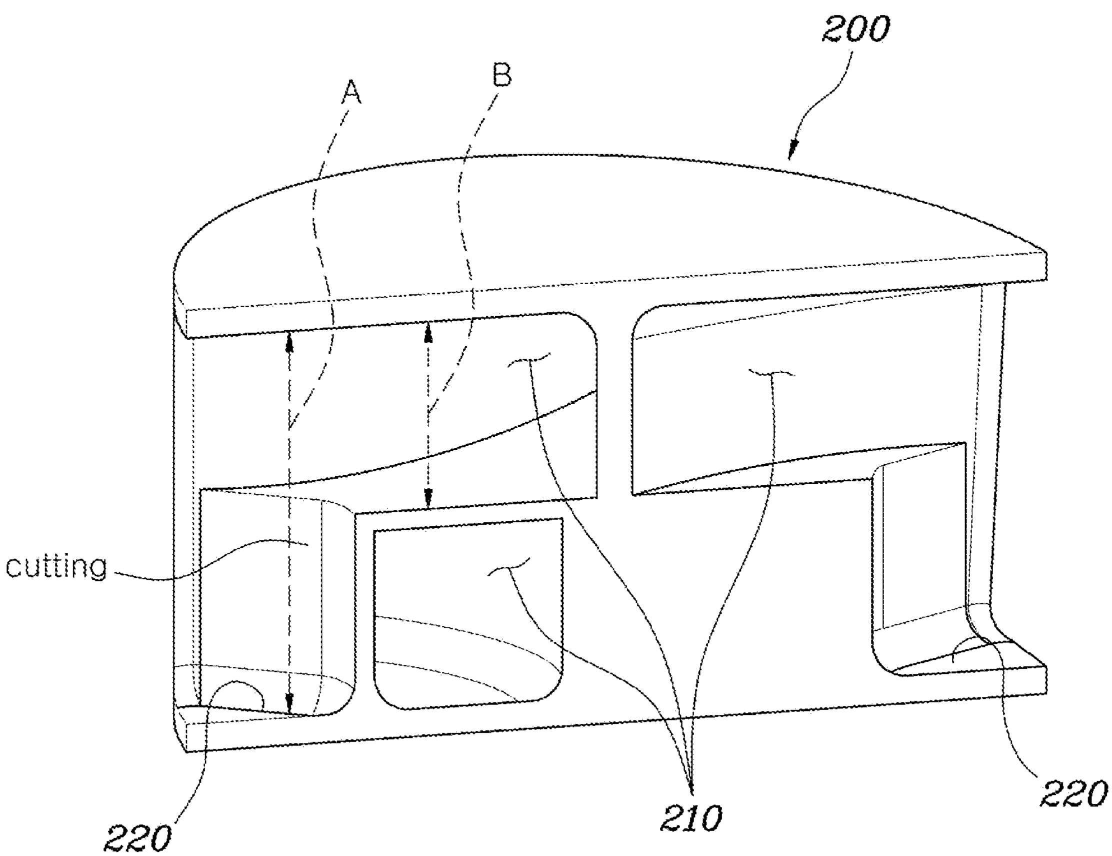


FIG. 7

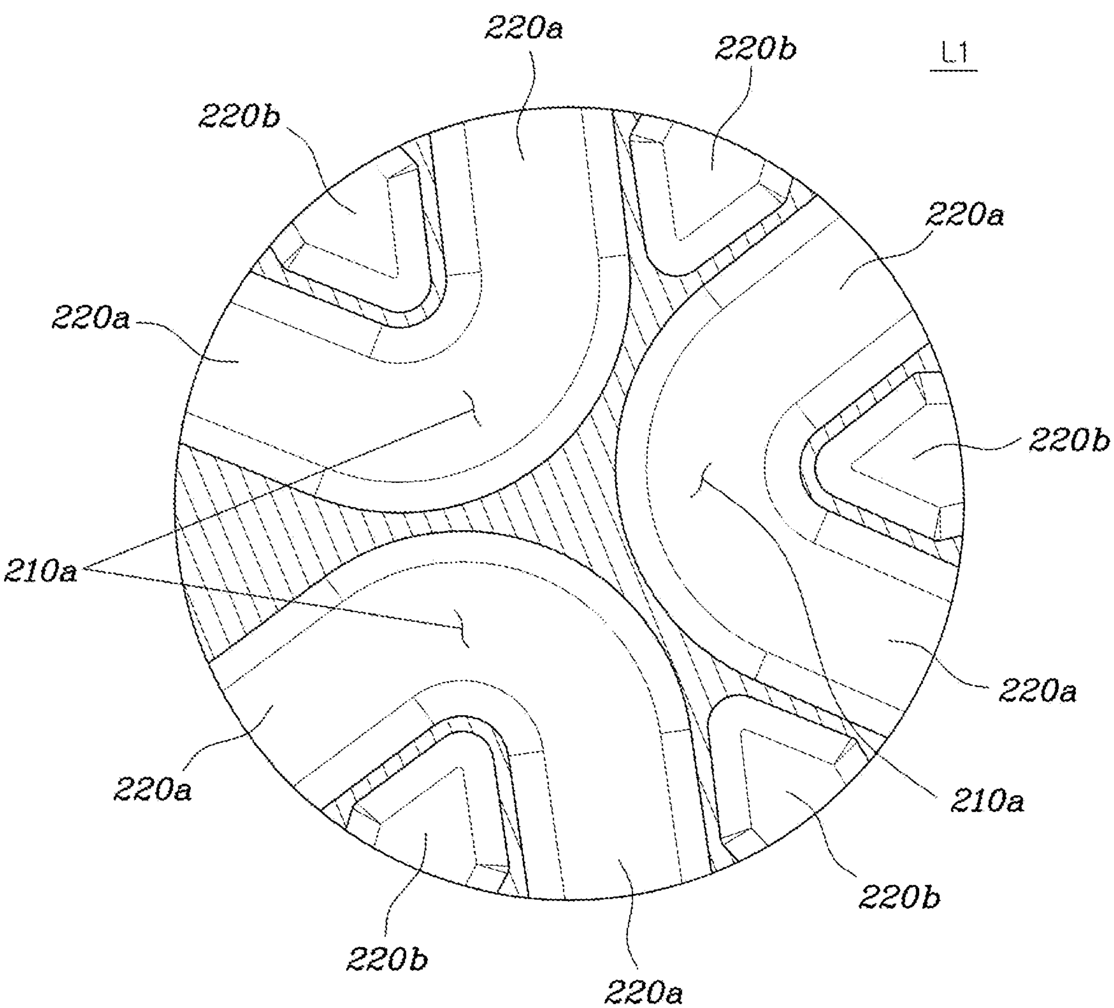

220a
220b
L1
220b
220a
220a
220b
210a
220a
220a
210a
220b
220b
220a

FIG. 8

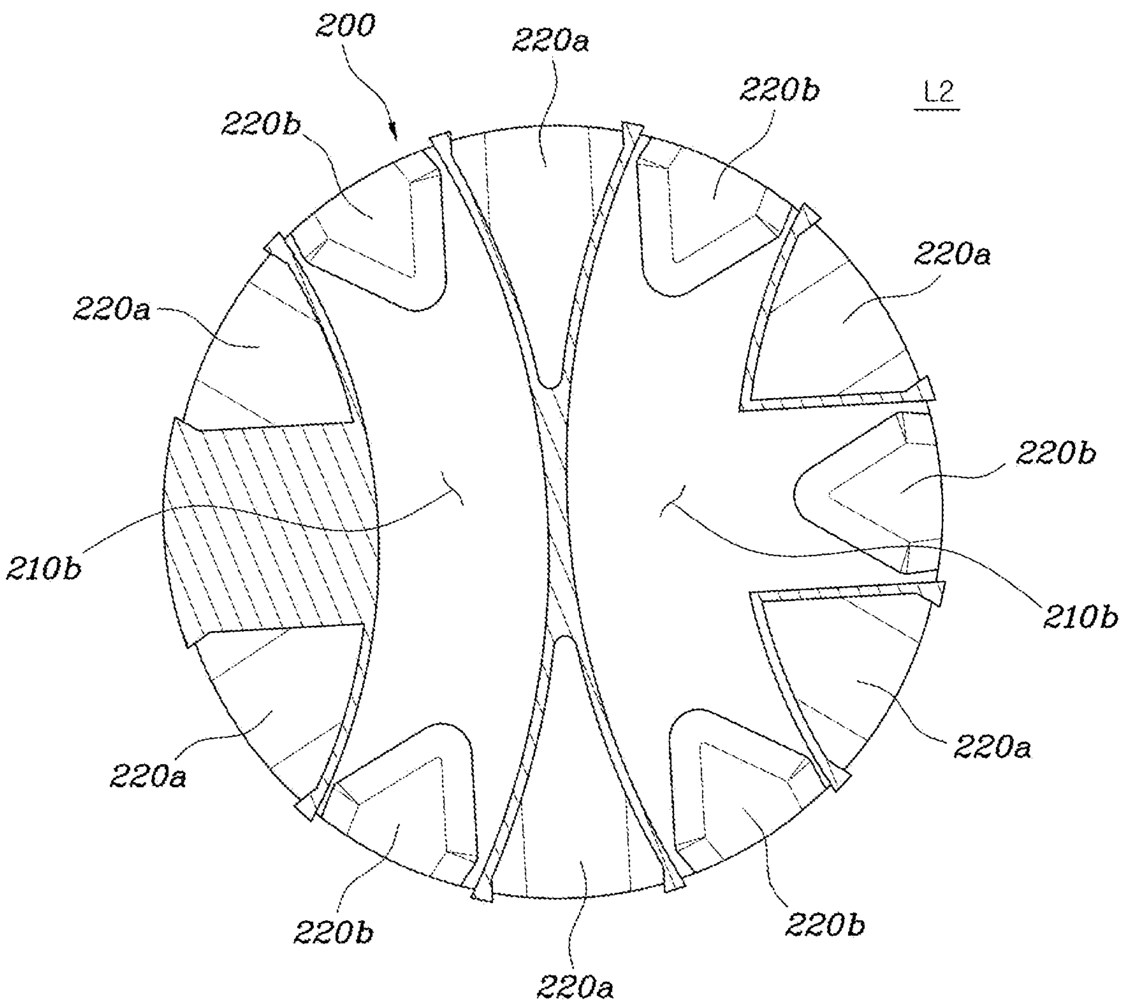

200
220a
220b
220b
L2
220a
220a
210b
220b
210b
220a
220a
220b
220b
220a

FIG. 9

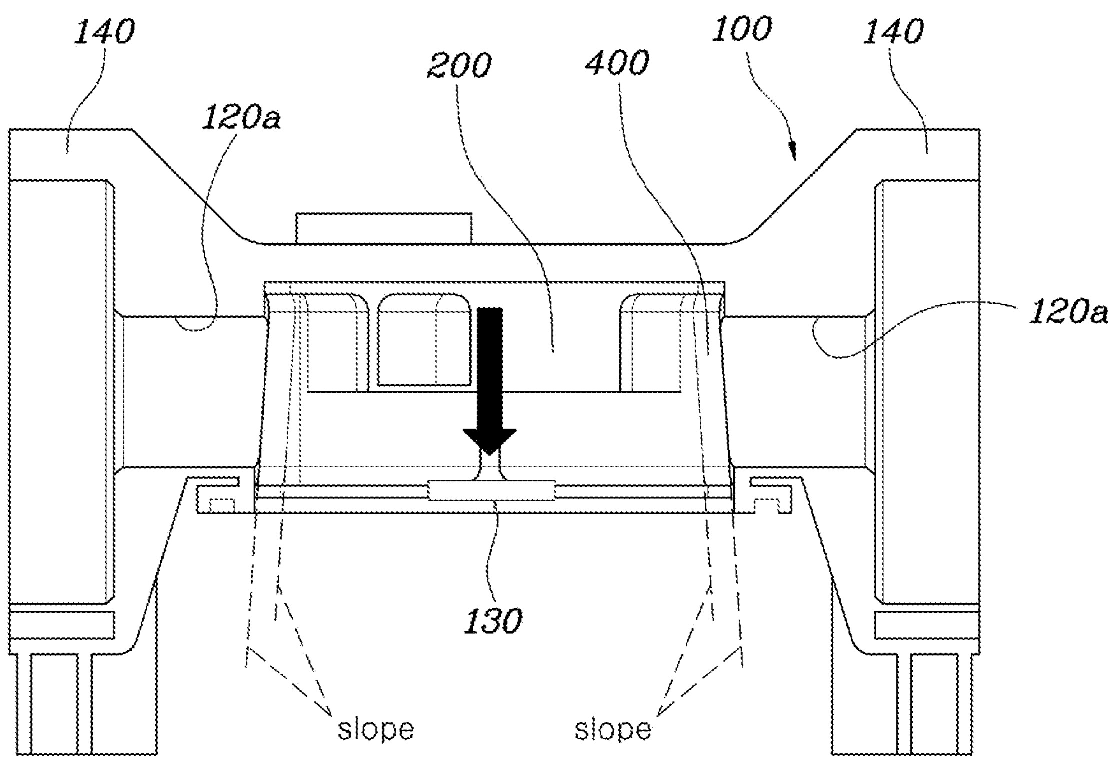


FIG. 10

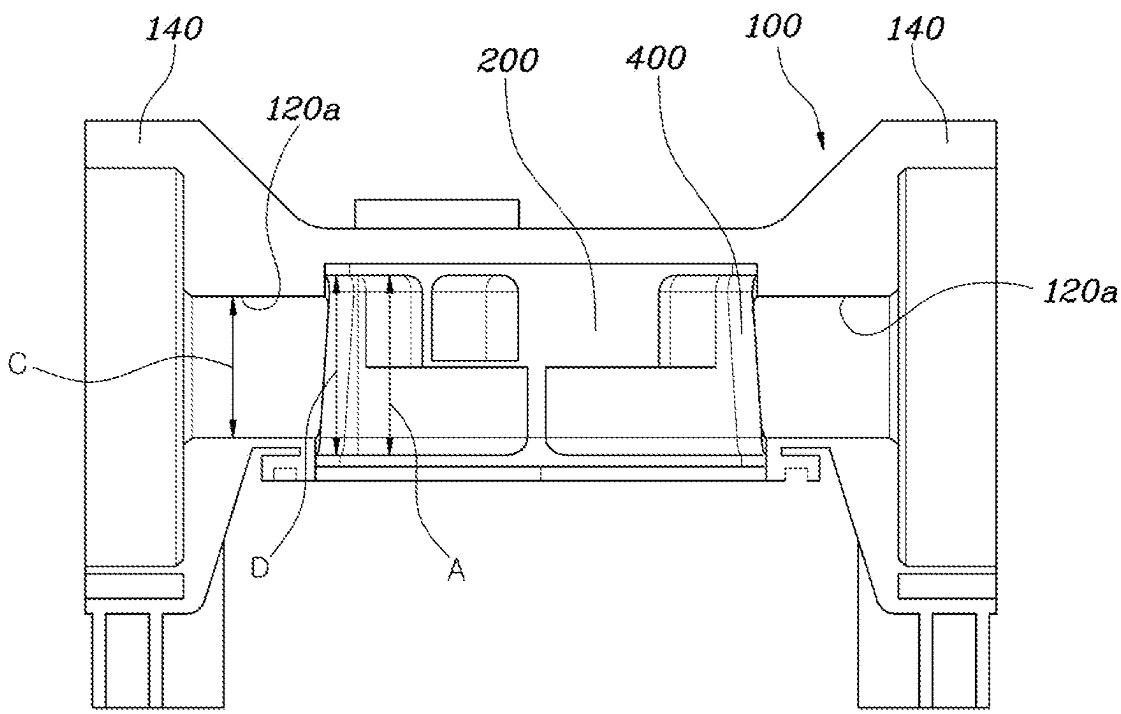


FIG. 11

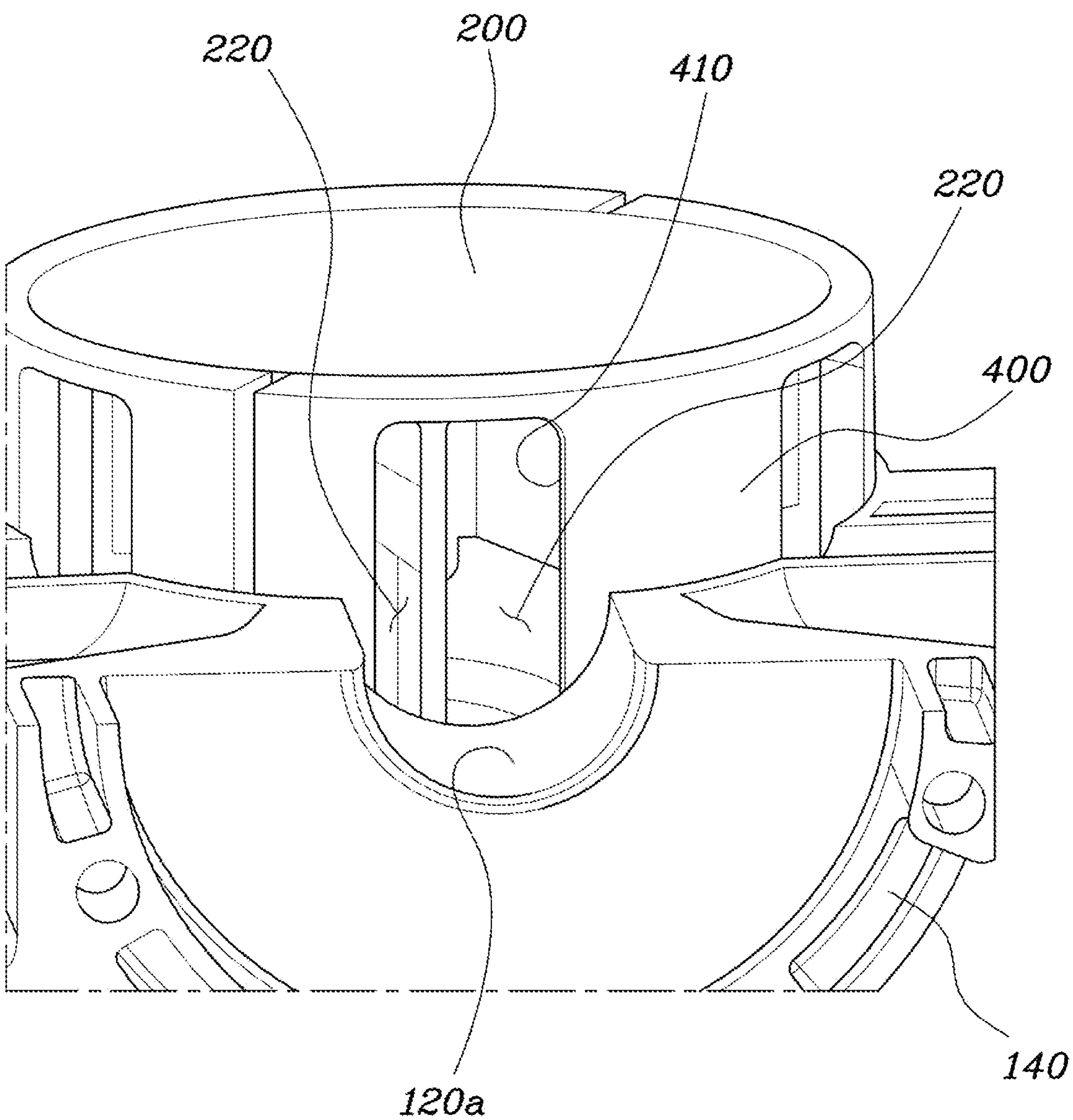
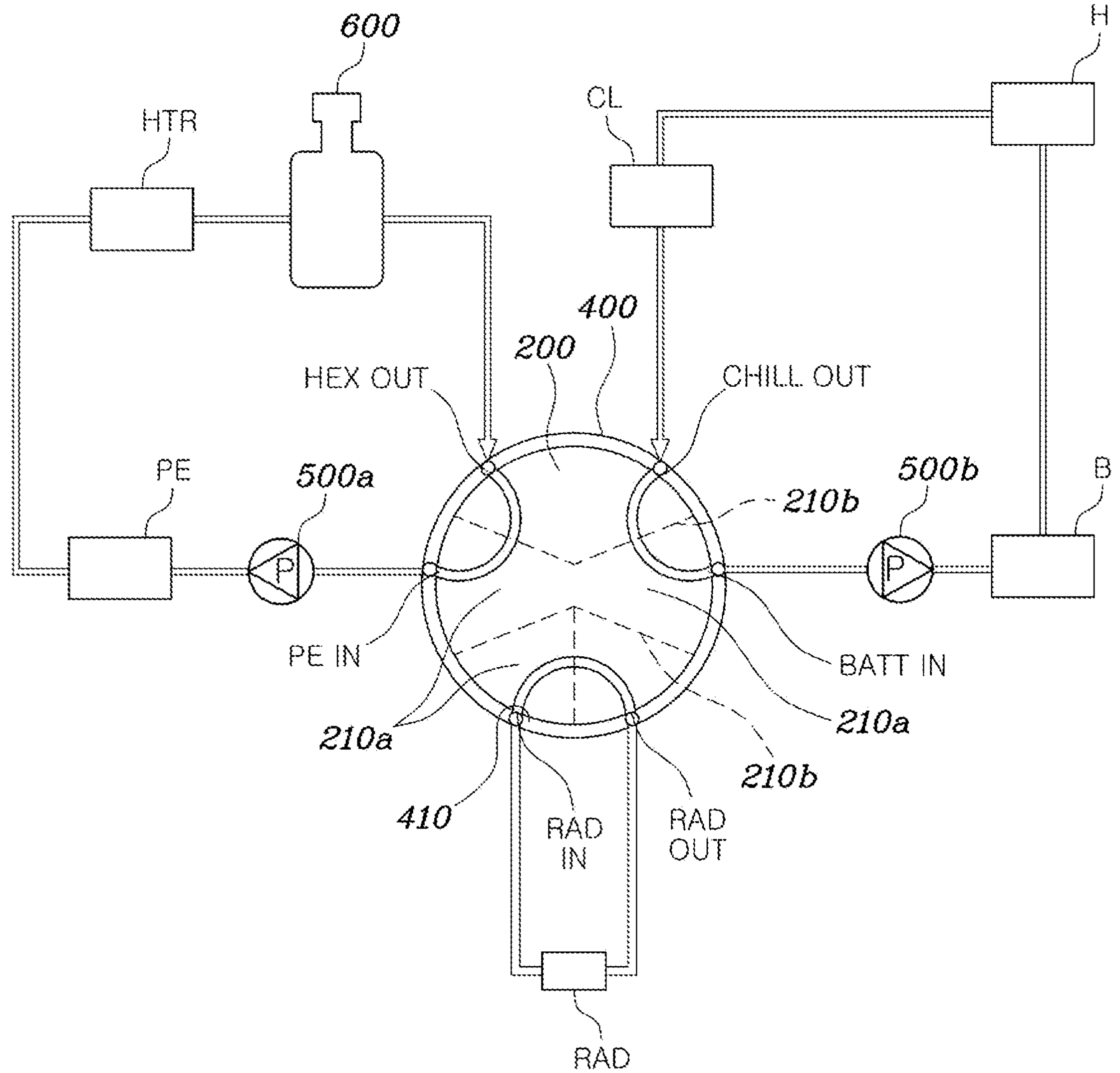


FIG. 12


600
HTR
CL
H
400
200
HEX OUT
CHILL OUT
PE
500a
210b
500b
B
P
P
PE IN
BATT IN
210a
210a
210b
410
RAD IN
RAD OUT
RAD

COOLING MEDIUM CIRCULATION APPARATUS FOR VEHICLE

CROSS REFERENCE TO RELATED APPLICATION

The present application claims under 35 U.S.C. § 119(a) the benefit of Korean Patent Application No. 10-2023-0001461, filed Jan. 4, 2023, the entire contents of which are incorporated by reference herein.

BACKGROUND

(a) Technical Field

The present disclosure relates to a cooling medium circulation apparatus for a vehicle, more particularly, to the cooling medium circulation apparatus capable of being installed in the vehicle in a compact structure, while eliminating the need for multiple peripheral pipes, and facilitating the circulation of coolant across various cooling system components.

(b) Description of the Related Art

In recent years, the increasing prevalence of eco-friendly vehicles, such as electric vehicles and fuel cell vehicles, has spurred significant advancements in related technologies. For eco-friendly vehicles reliant on electric energy from batteries or the like, technical advancements in energy efficiency are essential.

In the case of eco-friendly vehicles without an engine as a heat source, thermal management of the vehicle through the use of electrical energy is important to overall thermal efficiency.

The thermal management of eco-friendly vehicles can be divided into a battery, an electronic device, and interior air conditioning segments. In particular, it is more efficient to manage these segments in an integrated system than in separate systems, as this allows for the active utilization of waste heat and the enhancement of the overall energy consumption efficiency of the vehicle.

Such an integrated thermal management system can be designed along with developing and implementing the necessary components in a way that saves space and reduces weight within the vehicle, making it possible to manufacture more efficient vehicles.

The related art described above is intended merely to aid in the understanding of the background of the present disclosure, and should not be construed as recognizing the prior art that is known to those skilled in the art.

SUMMARY

It is an object of the present disclosure to provide a cooling medium circulation apparatus capable of being installed in vehicles in a compact structure, eliminating the need of multiple peripheral pipes, and facilitating the circulation of coolant across various cooling system components.

In order to accomplish the above objects, a cooling medium circulation apparatus for a vehicle according to an embodiment of the present disclosure includes a housing comprising a valve mounting device with an internal space and a plurality of ports connected to a plurality of thermal management components, and a valve body rotatably mounted in the valve mounting device, partitioned into a plurality of layers vertically, with a plurality of flow paths

formed in each layer, and with a plurality of circulation holes formed on outer sides corresponding to each of the flow paths and selectively matching the ports, the circulation holes elongated vertically to have a height greater than the height of the flow paths.

The valve mounting device is formed to have an inner surface forming a cylindrical shape, and the valve body is formed to have an outer surface forming a cylindrical shape facing the inner surface of the valve mounting device.

The cooling medium circulation apparatus further include an actuator connected to an upper or lower part of the housing to rotate the valve body to selectively establish a connection between the circulation holes of the valve body and the ports of the housing.

The plurality of ports are arranged at intervals on the outer surface of the valve mounting device.

The flow paths formed in one layer are separated from the flow paths formed in another layer.

The flow paths formed in one layer are vertically overlapped with the flow paths formed in another layer.

The circulation holes are cut to be recessed inward at a portion connected to the flow paths.

The cooling medium circulation apparatus further includes a sealing member disposed between the inner surface of the valve mounting device and the outer surface of the valve body, wherein the sealing member includes through-holes corresponding to the circulation holes.

The sealing member is cylindrical and is fixed in position when inserted into the valve mounting device, allowing the valve body to rotate relative to the sealing member.

The sealing member is provided with the through-holes in multiple quantities to match the quantities of the ports.

The ports of the housing are arranged at regular intervals, and the through-holes of the sealing member are arranged to match each port of the housing, allowing the distance between the through-holes to be also regularly spaced.

The circulation holes of the flow paths formed in one layer of the valve body and the circulation holes of the flow paths formed in another layer are alternately arranged along the outer surface of the valve body, and the through-holes within the sealing member are formed in fewer quantities than the circulation holes of the valve body and are arranged at intervals to match the circulation holes of the flow paths formed in one layer or the flow paths formed in another layer of the valve body, depending on the rotational position of the valve body.

The valve mounting device is open at the top or bottom, with a sloped inner surface that gradually widens in the open direction.

The valve body is formed with a sloped outer surface matching the sloped inner surface of the valve mounting device.

The housing includes a contact member having elastic and abrasion-resistant properties at the open portion of the valve mounting device to support the valve body.

The thermal management components comprise a water pump, and the housing comprises a mounting portion formed on a side thereof for attaching the water pump, and the mounting portion comprises a water pump port communicating with the water pump.

The water pump port has a height smaller than the height of the through-hole of the valve body.

The water pump port is positioned at the center of the circulation hole of the valve body.

The cooling medium circulation apparatus further includes a sealing member disposed between an inner surface of the valve mounting device and an outer surface of the valve body, wherein the sealing member comprises through-holes corresponding to the circulation holes and having a height larger than the height of the water pump port and smaller or equal to the height of the circulation hole.

The plurality of circulation holes are arranged, while one circulation hole is in an open state matched with the water pump port during the rotation of the valve body, for another circulation hole to be positioned to match the same water pump port and switch to the open state before the closure of the one circulation hole.

The cooling medium circulation apparatus further includes a reservoir attached to an upper part or lower part of the housing and storing the cooling medium.

The cooling medium circulation apparatus constructed as described above is advantageous in terms of being installed in a vehicle in a compact structure, eliminating the need of multiple peripheral pipes, and facilitating the circulation of coolant across various cooling system components.

Particularly, arranging the thermal management components such as the reservoir and water pumper centrally around the housing, allowing selective coolant circulation to each thermal management component through a signal valve and connection through each port without branch pipes offers advantages in terms of modularity for the cooling system components.

A vehicle may include a cooling medium circulation apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram illustrating a cooling medium circulation apparatus according to an embodiment of the present disclosure;

FIG. **2** is an exploded view of the cooling medium circulation apparatus of FIG. **1**;

FIG. **3** is a cross-sectional view illustrating a plurality of flow paths and valve bodies of the cooling medium circulation apparatus of FIG. **1**;

FIG. **4** is a diagram illustrating a valve body of the cooling medium circulation apparatus of FIG. **1**;

FIG. **5** is a cross-sectional view illustrating layers of the valve body of the cooling medium circulation apparatus of FIG. **1**;

FIG. **6** is a vertical cross-sectional view of the body valve of FIG. **5**;

FIG. **7** is a cross-sectional view in accordance with a first layer of the valve body of FIG. **5**;

FIG. **8** is a cross-sectional view in accordance with a second layer of the valve body of FIG. **5**;

FIG. **9** is a diagram illustrating another embodiment of the cooling medium circulation apparatus of FIG. **1**;

FIG. **10** is a diagram illustrating a water pump port of the cooling medium circulation apparatus of FIG. **1**;

FIG. **11** is a diagram illustrating another embodiment of the cooling medium circulation apparatus of FIG. **1**; and

FIG. **12** is a diagram illustrating a configuration of an integrated thermal management system employed a cooling medium circulation apparatus according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

It is understood that the term "vehicle" or "vehicular" or other similar term as used herein is inclusive of motor vehicles in general such as passenger automobiles including sports utility vehicles (SUV), buses, trucks, various commercial vehicles, watercraft including a variety of boats and ships, aircraft, and the like, and includes hybrid vehicles, electric vehicles, plug-in hybrid electric vehicles, hydrogen-powered vehicles and other alternative fuel vehicles (e.g. fuels derived from resources other than petroleum). As referred to herein, a hybrid vehicle is a vehicle that has two or more sources of power, for example both gasoline-powered and electric-powered vehicles.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. Throughout the specification, unless explicitly described to the contrary, the word "comprise" and variations such as "comprises" or "comprising" will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. In addition, the terms "unit", "-er", "-or", "portion" and "module" described in the specification mean units for processing at least one function and operation, and can be implemented by hardware components or software components and combinations thereof.

Further, the control logic of the present disclosure may be embodied as non-transitory computer readable media on a computer readable medium containing executable program instructions executed by a processor, controller or the like. Examples of computer readable media include, but are not limited to, ROM, RAM, compact disc (CD)-ROMs, magnetic tapes, floppy disks, flash drives, smart cards and optical data storage devices. The computer readable medium can also be distributed in network coupled computer systems so that the computer readable media is stored and executed in a distributed fashion, e.g., by a telematics server or a Controller Area Network (CAN).

Hereinafter, descriptions are made of the embodiments disclosed in the present specification with reference to the accompanying drawings in which the same reference numbers are assigned to refer to the same or like components and redundant description thereof is omitted.

In addition, detailed descriptions of well-known technologies related to the embodiments disclosed in the present specification may be omitted to avoid obscuring the subject matter of the embodiments disclosed in the present specification. In addition, the accompanying drawings are only for easy understanding of the embodiments disclosed in the present specification and do not limit the technical spirit disclosed herein, and it should be understood that the embodiments include all changes, equivalents, and substitutes within the spirit and scope of the disclosure.

As used herein, terms including an ordinal number such as "first" and "second" can be used to describe various components without limiting the components. The terms are used only for distinguishing one component from another component.

It will be understood that when a component is referred to as being "connected to" or "coupled to" another component, it can be directly connected or coupled to the other component or intervening component may be present. In contrast, when a component is referred to as being "directly connected to" or "directly coupled to" another component, there are no intervening component present.

As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

Hereinafter, a description is made of the cooling medium circulation apparatus according to a preferred embodiment of the present disclosure with reference to accompanying drawings.

FIG. **1** is a diagram illustrating a cooling medium circulation apparatus according to an embodiment of the present disclosure, FIG. **2** is an exploded view of the cooling medium circulation apparatus of FIG. **1**, FIG. **3** is a cross-sectional view illustrating a plurality of flow paths and valve bodies of the cooling medium circulation apparatus of FIG. **1**, FIG. **4** is a diagram illustrating a valve body of the cooling medium circulation apparatus of FIG. **1**, FIG. **5** is a cross-sectional view illustrating layers of the valve body of the cooling medium circulation apparatus of FIG. **1**, FIG. **6** is a vertical cross-sectional view of the body valve of FIG. **5**, FIG. **7** is a cross-sectional view in accordance with a first layer of the valve body of FIG. **5**, and FIG. **8** is a cross-sectional view in accordance with a second layer of the valve body of FIG. **5**.

FIG. **9** is a diagram illustrating another embodiment of the cooling medium circulation apparatus of FIG. **1**, FIG. **10** is a diagram illustrating a water pump port of the cooling medium circulation apparatus of FIG. **1**, and FIG. **11** is a diagram illustrating another embodiment of the cooling medium circulation apparatus of FIG. **1**.

FIG. **12** is a diagram illustrating a configuration of an integrated thermal management system employed a cooling medium circulation apparatus according to an embodiment of the present disclosure.

As shown in FIGS. **1** to **8**, the cooling medium circulation apparatus according to an embodiment of the present disclosure includes a housing **100** equipped with a valve mounting device **110** with an internal space and ports **120** connected to a plurality of thermal management components, and a valve body **200** that is rotatably mounted in the valve mounting device **110**, partitioned into a plurality of layers vertically, with a plurality of flow paths **210** formed in each layer, and with circulation holes **220** formed on the outer sides corresponding to each flow path **210** and selectively matching the ports **120**, the circulation holes **220** elongated vertically to have a height A greater than the height B of the flow paths **210**.

The housing **100** is provided with the valve mounting device **110** in order for the valve body **200** is rotatably mounted inside the valve mounting device **110**. By forming the valve mounting device **110** inside the housing **100**, the structure may become more compact as the valve body **200** is positioned within the housing **100**.

An actuator **300** and a reservoir **600** may be coupled to the upper or lower part of the housing **100**. The reservoir **600** is configured to store the cooling medium, while the actuator **300** is connected to the valve body **200**, allowing for the rotation of the valve body **200**.

According to an embodiment of the present disclosure, the housing **100** is configured to smoothly circulate the colling medium stored in the reservoir connected at the upper part thereof and to ensure no interference between the actuator **300** connected at the lower part thereof and other components positioned within the housing **100**.

In this way, the reservoir **600** and actuator **300** are mounted on the upper and lower parts of the housing **100**, respectively, and the water pump **500** to be described later is mounted on the side, thereby reducing the overall package.

The actuator **300** may be controlled by a controller **700**, and the controller **700** may be mounted on the reservoir **600**.

The valve mounting device **110** of the housing **100** is equipped with a plurality of ports **120**, which may be arranged on the side of the housing **100** to be spaced apart from each other.

Here, the plurality of ports **120** are arranged along the circumference of the valve mounting device **110**, facilitating the connection of the cooling medium circulation line to the ports **120** and preventing interference with other components.

Meanwhile, the valve mounting device **110** is designed to allow the valve body **200** to rotate, and depending on the rotational position of the valve body **200**, the circulation direction of the cooling medium through the flow paths **210** is switched by matching the circulation holes **220** to the ports **120**.

In addition, the inner surface of the valve mounting device **110** and the outer surface of the valve body **200** are cylindrically shaped, allowing the outer surface of the valve body **200** to face the inner surface of the valve mounting device **110**.

That is, the valve mounting device **110** may be formed in a cylindrical shape recessed into the inside of the housing **100**. The valve mounting device **110**, designed as such, accommodates the cylindrical valve body **200** to rotate coaxially.

The plurality of ports **120** of the valve mounting device **110** and the plurality of circulation holes **220** of the valve body **200** may be selectively matched depending on the rotational position of the valve body **200**, allowing the cooling medium to circulate through each thermal management component. This makes it possible to implement various thermal management modes by diversifying the circulation direction of the cooling medium depending on the rotational position of the valve body **200**.

Meanwhile, in the present disclosure, the valve body **200** is divided into a plurality of layers vertically, with a plurality of flow paths **210** formed in each layer. That is, the valve body **200** has a plurality of flow paths **210** formed in each layer, and each flow path **210** may have a plurality of circulation holes **220** serving as the inlet and outlet.

Accordingly, the valve body **200** has multiple flow paths **210** configured for each layer, resulting in a greater number of circulation holes **220** compared to the number of flow paths **210**, each serving as the inlet and outlet for the flow paths **210**.

The circulation holes **220** are arranged at intervals along the outer surface of the valve body **200** and are each elongated vertically to have a height A longer than the height B of the flow paths **210**. As a result, the cooling medium may flow through the flow paths **210** via the circulation holes **220** of the valve body **200**, ensuring a sufficient flow rate. Furthermore, separating the circulation for each flow path **210** via the plurality of circulation holes **220** can make it possible to precisely control the circulation direction of the cooling medium.

An embodiment of the present disclosure is directed to an exemplary configuration with two layers, the first layer L1 provided with a total of 3 flow paths **210** and 6 circulation holes **220** corresponding to each flow path **210** and the second layer L2 provided with a total of 2 flow paths **210** and 5 circulation holes **220** corresponding to each flow path **210**. In order to aid in understanding the disclosure, the flow paths **210***a* and circulation holes **220***a* corresponding to the first layer L1 and the flow paths **210***b* and circulation holes **220***b* corresponding to the second layer L2 are illustrated separately in FIGS. **7** and **8**.

These layers may be provided in various forms without being limited to this embodiment.

In particular, the flow paths **210** formed in one layer may be separated from the flow paths **210** formed in another layer in the present disclosure.

The flow paths **210** formed in one layer may also be vertically or horizontally overlapped with the flow paths **210** formed in another layer.

That is, the valve body **200** may be partitioned into a plurality of layers formed each to have the flow paths **210**.

Here, the flow paths **210** are formed to be arranged at intervals from each other within the first layer L1 of the valve body **200**.

The flow paths **210** of the second layer L2 are formed to be vertically separated from the flow paths **210** of the first layer L1 and arranged at intervals from each other within the second layer L2. This arrangement allows the flow paths **210** of the second layer L2 to be vertically overlapped with the flow paths **210** of the first layer L1 and aligned on a vertical line together, thereby reducing the overall size of the valve body **200**.

The flow paths **210** of the first and second layers L1 and L2 are positioned and oriented differently within the valve body **200**, allowing for the rotation of the valve body **200** to form cooling medium flows in various directions.

The circulation holes **220** on the outer surface of the valve body **200** are spaced apart from each other to correspond to the flow paths **210** of the first and second layers L1 and L2, allowing for distinction between the inlet and outlet of each flow path **210**.

Meanwhile, the circulation holes **220** may be cut to be recessed inward at a portion connected to the flow paths **210**.

As shown in FIG. **6**, the circulation holes **220** are cut to be recessed inward at the portion connected to the flow paths **210**, allowing for the smoothly entry of the cooling medium into the flow paths **210** through the circulation holes **220**. Here, the portion where the circulation holes **220** connect to the flow paths **210** may be V-cut, and the cutting shape may be applied in various forms.

Meanwhile, a sealing member **400** may be interposed between the inner surface of the valve mounting device **110** and the outer surface of the valve body **200**, and the sealing member **400** may have through-holes **410** corresponding to the circulation holes **220**.

The sealing member **400** is cylindrical and is fixed in position when inserted into the valve mounting device **110**, allowing the valve body **200** to rotate relative to the sealing member **400**.

The through-holes **410** of the sealing member **400** may be formed in the same number as the ports **120** of the housing **100** and may be shaped to match the circulation holes **220** of the valve body **200**.

Accordingly, the sealing member **400** ensures a sealed configuration, except for the through-holes **410**, allowing the circulation holes **220** matched with the through-holes **410** to facilitate the flow of the cooling medium, while blocking the circulation holes **220** unmatched with through-holes **410** to block the circulation of the cooling medium, depending on the rotational position of the valve body **200**.

The sealing member **400** may be provided with the through-holes **410** in multiple quantities to match multiple ports **120**.

Here, the ports **120** of the housing **100** may be arranged at regular intervals, and the through-holes **410** of the sealing member **400** may be formed in an equal number to match the ports **120**, or in a greater number.

In the case of being formed in the equal number, the through-holes **410** of the sealing member **400** may be arranged at the same intervals as the ports **120**, ensuring a precise alignment between the through-holes **410** and the respective ports **120**.

In the case of being formed in the greater number, the through-holes **410** of the sealing member **400** may each be arranged to match each port **120**, with some through-holes **410** redundantly matching one port **120**. This allows for diversifying the circulation of the cooling medium for each port **120** depending on the rotational position of the valve body **200**.

According to an embodiment of the present disclosure, the ports **120** of the housing **100** are arranged at regular intervals, and accordingly, the through-holes **410** of the sealing member **400** are formed to match the number and spacing of the ports **120**, as shown in FIG. **3**. As a result, the sealing member **400** is arranged with the remaining parts, excluding the through-holes **410**, at equidistant intervals between the inner surface of the valve mounting device **110** and the outer surface of the valve body **200**, allowing the circumferential surface to maintain a uniform contact state with balanced repulsive force between the valve mounting device **110** and the valve body **200**.

Also, the valve body **200** may have the circulation holes **220** of the flow paths **210** formed in one layer and the circulation holes **220** of the flow paths **210** formed in another layer alternately arranged along the outer side, and the sealing member **400** may have the through-holes **410** fewer in number than the circulation holes **220** of the valve body **200** and arranged at intervals to match either the circulation holes **220** of the flow paths **210** formed in one layer or the circulation holes **220** of the flow paths **210** formed in another layer depending on the rotational position of the valve body **200**.

That is, the circulation holes **220** of the valve body **200** may be distinguished by each flow path **210** formed in each layer, and the circulation holes **220** corresponding to the flow paths **210** formed in each layer may be alternately arranged at regular intervals.

The through-holes **410** of the sealing member **400** may be formed in the same quantity as the number of ports **120** and thus being fewer in number than the circulation holes **220** of the valve body **200**. In particular, the plurality of through-holes **410** are formed to be spaced apart from each other to match the circulation holes **220** of the flow paths **210** formed in one of the plurality of layers, and depending on the rotational position of the valve body **200**, the through-holes **410** may be aligned with the circulation holes **220** of the flow paths **210** formed in one layer or the circulation holes **220** of the flow paths **210** formed in another layer.

For example, FIG. **3** shows the second layer L2 among the plurality of layers in the case where the valve body **200** is rotated to align the circulation hole **220***b* corresponding to the second layer L2 with the through-hole **410** of the sealing member **400**, which aligns the circulation hole **220***b* of the second layer L2, the through-hole **410** of the sealing member **400**, and the port **120** of the housing **100**, allowing for the circulation of the cooling fluid. On the other hand, the circulation holes **220***a* corresponding to the first layer L1 are closed off by matching with the body surface of the sealing member **400**, preventing the circulation of the cooling fluid. As described above, the present disclosure may be configured to allow the cooling fluid to flow through either the circulation holes **220***a* corresponding to the first layer L1 or the circulation holes **220***b* corresponding to the second layer L2 of the plurality of layers, depending on the rotational position of the valve body **200**, as described above.

Meanwhile, as shown in FIG. **9**, the valve mounting device **110** may be open at the top or bottom, with a sloped inner surface that gradually widens towards the opening.

In addition, the valve body **200** may be formed to have an outer surface sloped to match the inner surface of the valve mounting device **110**.

In this way, the inner surface of the valve mounting device **110** and the outer surface of the valve body **200** are formed with a slope, allowing the valve body **200** to move in the direction of the opening of valve mounting device **110** by the force generated during the circulation of the cooling fluid.

The valve mounting device **110** has a sloped inner surface that gradually widens in the direction of the opening, and the valve body **200** is also formed with a sloped outer surface matching the sloped inner surface of the valve mounting device **110**, thereby reducing friction against the valve mounting device **110** as the valve body **200** moves in the direction of the opening of the valve mounting device **110** under the force of the circulating medium. That is, the tapered shape applied to both the valve mounting device **110** and the valve body **200** is capable of preventing potential damage by reducing the friction generated on the valve mounting device **110** as the valve body **200** rotates in the state where the valve body **200** inserted into the valve mounting device **110** is in contact with the valve mounting device **110**.

Here, the housing **100** is provided with a contact member **130** having elastic and abrasion-resistant properties at the open portion of the valve mounting device **110**, thereby allowing the valve body **200** to be supported by the contact member **130**. The contact member **130** is provided at the open portion of the valve mounting device **110** to support the valve body **200** and may also be provided in the actuator **300**. The contact member **130** may be composed of a material with elasticity and abrasion resistance to facilitate the rotation of the valve body **200** and minimize damage due to friction.

As a result, the valve body **200** can ensure operational performance and durability when inserted into the valve mounting device **110**.

Meanwhile, as shown in FIG. **2** and FIG. **10**, the thermal management components include a water pump **500**, and the housing **100** may be provided with a mounting portion **140** on its side for mounting the water pump **500**, the mounting portion **140** having a water pump port **120***a* communicating with the water pump **500**.

In an embodiment of the present disclosure, the water pump **500** is mounted on the side of the housing **100**, and a mounting portion **140** is formed on the housing **100** for mounting the water pump **500**. The mounting portion **140** of the housing **100** may be formed to match the outer shape of the water pump **500** to allow the water pump **500** to be attached, and the water pump port **120***a* may be formed at the center, allowing the cooling medium propelled by the water pump **500** to flow through the water pump port **120***a* into the valve body **200**. Here, the water pump port **120***a* may be implemented in the form of a hole.

Here, the height C of the water pump port **120***a* may be formed to be smaller than the height A of the circulation hole **220** of the valve body **200**.

In the present disclosure, the direct connection of the water pump **500** to the housing **100** necessitates the improvement of the circulation flow of the cooling medium during the operation of the water pump **500**. Accordingly, it is necessary to design the water pump port **120***a* with a height C smaller than the height A of the circulation hole **220** in the valve body **200** to prevent any blockage of the cooling medium caused by the reduction in the flow path **210**, stabilizing the circulation flow between the water pump **500** and the valve body **200**.

Along with this, positioning the water pump port **120***a* at the center of the circulation hole **220** of the valve body **200** minimizes the reduction in the flow path **210** during the matching between the water pump port **120***a* and the circulation hole **220** of the valve body **200**, preventing interference in the flow of the cooling medium and enhancing the flow characteristics of the cooling medium.

In addition, the height D of the through-hole **410** is larger than the height C of the water pump port **120***a* and smaller than or equal to the height A of the circulation hole **220**.

Thus, forming the through-hole **410** of the sealing member **400**, positioned between the inner surface of the valve mounting device **110** and the outer surface of the valve body **200**, to have the height D equal to or less than the height A of the circulation hole **220** of the valve body **200** may allow the valve body **200** to maintain a closed state even if there is any misalignment or offset.

Additionally, as the height D of the through-hole **410** is greater than the height C of the water pump port **120***a*, it becomes possible to prevent a decrease in the flowability of the cooling medium caused by the reduction in the flow path **210**.

In this way, by regulating the sizes of the water pump port **120***a* of the housing **100**, the circulation hole **220** of the valve body **200**, and the through-hole **410** of the sealing member **400** during the installation of the water pump **500** on the housing **100**, it is possible to improve the flowability of the cooling medium and ensure reliability against any blockages of the cooling medium.

Meanwhile, as shown in FIG. **11**, the plurality of circulation holes **220** may be arranged such that while one circulation hole **220** is in an open state matched with the water pump port **120***a* during the rotation of the valve body **200**, another circulation hole **220** can be positioned to match the same water pump port **120***a* and switch to the open state before the closure of the first circulation hole **220**.

In this way, the plurality of circulation holes **220** may be arranged adjacent to each other on the outer surface of the valve body **200** through the arrangement of the flow paths **210** of each layer.

That is, the position change of the valve body **200** may block the circulation of the cooling medium through the water pump port **120***a*, resulting in momentary high load on the water pump **500** and causing problems such as noise, cavitation, air bubbles, etc. Accordingly, in the present disclosure, during the rotation of the valve body **200**, while one of the plurality of circulation holes **220** is in an open state matching the water pump port **120***a*, another one of the circulation holes **220** transitions to the open state before the first circulation hole **220** transitions to the closed state, ensuring uninterrupted flow of the cooling medium, thereby improving the durability of the water pump **500** and minimizing noise and cavitation during the transition operation of the valve body **200**.

The present disclosure described above can diversify the flow of the cooling medium depending on the rotational position of the valve body **200**, allowing for the diversification of the cooling function according to the flow direction of the cooling medium.

FIG. **12** shows an integrated thermal management system employing a cooling medium circulation apparatus including the valve body **200** with 11 circulation holes **220** and the sealing member **400** with 6 through-holes **410** according to an embodiment of the present disclosure. In the drawing, the PE IN port is connected to the HEX OUT port through a first water pump **500***a*, a power electronics PE, a heat exchanger HTR, and a reservoir **600**. The RAD IN port is connected to the RAD OUT port through the radiator RAD. The BATT IN port is connected to the CHILL OUT port through a second water pump **500***b*, a battery B, a heater H, and a chiller CL. This connection setup allows for independent control of the cooling medium flow to the power electronics PE, radiator RAD, and battery B by rotating the valve body **200**, enabling various thermal management modes to be implemented by varying the flows of cooling medium through the flow paths **210** in the first and second layers L1 and L2.

The cooling medium circulation apparatus constructed as described above is advantageous in terms of being installed in a vehicle in a compact structure, eliminating the need of multiple peripheral pipes, and facilitating the circulation of coolant across various cooling system components.

Particularly, arranging the thermal management components such as the reservoir and water pumper centrally around the housing, allowing selective coolant circulation to each thermal management component through a single valve and connection through each port without branch pipes offers advantages in terms of modularity for the cooling system components.

Although the present disclosure has been illustrated and described in connection with specific embodiments, it will be obvious to those skilled in the art that various modification and changes can be made thereto without departing from the spirit of the disclosure or the scope of the appended claims.

What is claimed is:

1. A cooling medium circulation apparatus for a vehicle, the cooling medium circulation apparatus comprising:

a housing comprising a valve mounting device with an internal space and a plurality of ports connected to a plurality of thermal management components; and

a valve body rotatably mounted in the valve mounting device, partitioned into a plurality of layers vertically, with a plurality of flow paths formed in each layer, and with a plurality of circulation holes formed on an outer side corresponding to each of the flow paths and selectively matching the ports, the circulation holes elongated vertically to have a height greater than a height of the flow paths.

2. The cooling medium circulation apparatus of claim **1**, wherein the valve mounting device is formed to have an inner surface forming a cylindrical shape, and the valve body is formed to have an outer surface forming the cylindrical shape facing the inner surface of the valve mounting device.

3. The cooling medium circulation apparatus of claim **1**, further comprising an actuator connected to an upper or lower part of the housing to rotate the valve body to selectively establish a connection between the circulation holes of the valve body and the ports of the housing.

4. The cooling medium circulation apparatus of claim **1**, wherein the plurality of ports are arranged at intervals on an outer surface of the valve mounting device.

5. The cooling medium circulation apparatus of claim **1**, wherein the flow paths formed in one layer are separated from, or vertically overlapped with, the flow paths formed in another layer.

6. The cooling medium circulation apparatus of claim **1**, wherein the circulation holes are cut to be recessed inward at a portion connected to the flow paths.

7. The cooling medium circulation apparatus of claim **1**, further comprising a sealing member disposed between an inner surface of the valve mounting device and an outer surface of the valve body, the sealing member comprising through-holes corresponding to the circulation holes.

8. The cooling medium circulation apparatus of claim **7**, wherein the sealing member is cylindrical and is fixed in position when inserted into the valve mounting device, allowing the valve body to rotate relative to the sealing member.

9. The cooling medium circulation apparatus of claim **7**, wherein the sealing member is provided with the through-holes in multiple quantities to match the quantities of the ports.

10. The cooling medium circulation apparatus of claim **7**, wherein the ports of the housing are arranged at regular intervals, and the through-holes of the sealing member are arranged to match each port of the housing, allowing the distance between the through-holes to be also regularly spaced.

11. The cooling medium circulation apparatus of claim **7**, wherein the circulation holes of the flow paths formed in one layer of the valve body and the circulation holes of the flow paths formed in another layer are alternately arranged along the outer surface of the valve body, and the through-holes within the sealing member are formed in fewer quantities than the circulation holes of the valve body and are arranged at intervals to match the circulation holes of the flow paths formed in one layer or the flow paths formed in another layer of the valve body, depending on rotational position of the valve body.

12. The cooling medium circulation apparatus of claim **1**, wherein the valve mounting device is open at a top or a bottom thereof, with a sloped inner surface that gradually widens in an open direction.

13. The cooling medium circulation apparatus of claim **12**, wherein the valve body is formed with a sloped outer surface matching the sloped inner surface of the valve mounting device.

14. The cooling medium circulation apparatus of claim **12**, wherein the housing comprises a contact member having elastic and abrasion-resistant properties at the open portion of the valve mounting device to support the valve body.

15. The cooling medium circulation apparatus of claim **1**, wherein the thermal management components comprise a water pump, and the housing comprises a mounting portion formed on a side thereof for attaching the water pump, and the mounting portion comprises a water pump port communicating with the water pump.

16. The cooling medium circulation apparatus of claim **15**, wherein the water pump port has a height smaller than a height of a through-hole of the valve body, and the water pump port is positioned at a center of one of the circulation holes of the valve body.

17. The cooling medium circulation apparatus of claim **16**, further comprising a sealing member disposed between an inner surface of the valve mounting device and an outer surface of the valve body,

wherein the sealing member comprises through-holes corresponding to the circulation holes and having a height larger than the height of the water pump port and smaller or equal to the height of the circulation hole.

18. The cooling medium circulation apparatus of claim **16**, wherein the plurality of circulation holes are arranged, while one circulation hole is in an open state matched with the water pump port during the rotation of the valve body, for another circulation hole to be positioned to match the same water pump port and switch to the open state before the closure of the one circulation hole.

19. The cooling medium circulation apparatus of claim **1**, further comprising a reservoir attached to an upper part or lower part of the housing and storing the cooling medium.

20. A vehicle comprising the cooling medium circulation apparatus of claim **1**.

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